MALWARE ANALYSIS & FORENSICS

LAB REPORT

SUBMITTED BY

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M.SC. DIGITAL FORENSICS & INFORMATION SECURITY

SEMESTER II

SUBMITTED TO

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1. Malware Lab Setup — Virtual Environment

1.1 Objective

The objective of this laboratory exercise is to design, deploy, and validate a safe and fully isolated malware analysis environment using virtual machines. The lab enables controlled execution and observation of malware samples, monitoring of behavioral patterns, and capture of network activity — without exposing the host system or the external network infrastructure.

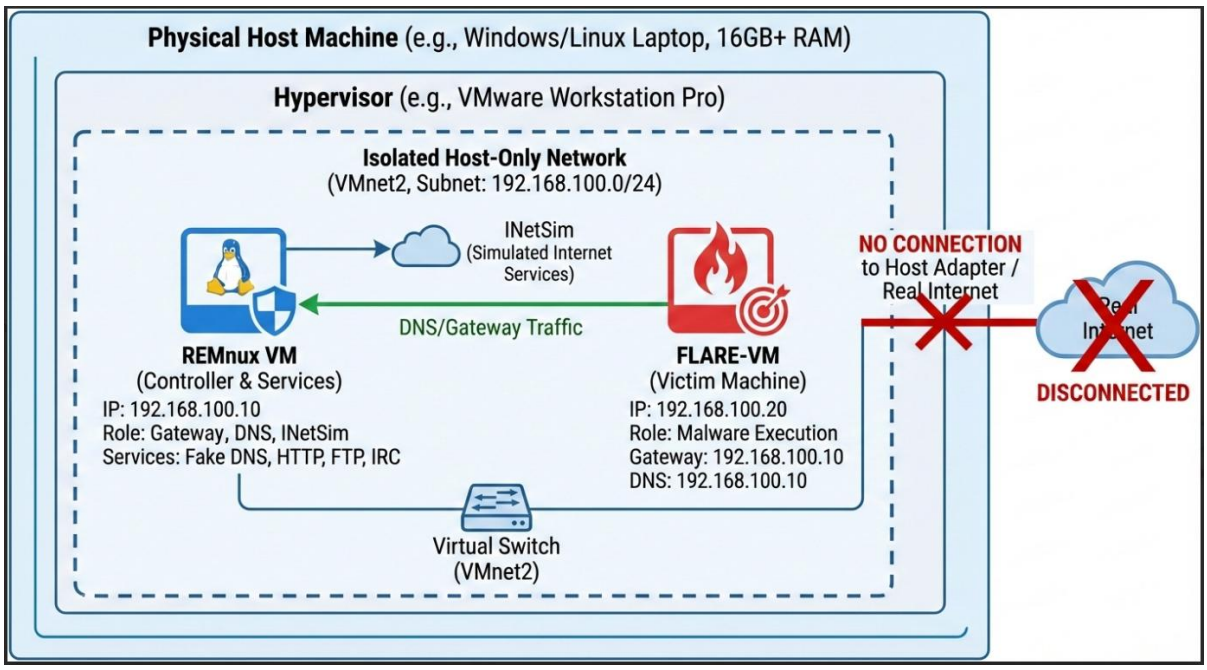


Figure 1.1 — Virtual Lab Environment Overview (VirtualBox with REMnux & FlareVM)

1.2 Lab Architecture

The analysis environment consists of three logically separated components operating within an air-gapped virtual network:

Component	Role	OS / Tool
Host System	Hypervisor — runs and manages VMs	Windows 11 / VirtualBox
Linux VM (REMnux)	Simulates internet services, captures network traffic, runs analysis tools	REMnux (Ubuntu-based)
Windows VM (FlareVM)	Executes malware samples in a controlled sandbox	FlareVM (Windows 10/11)

1.3 Configuration Steps

Step 1: VM Installation

- Installed VirtualBox on the host system.
- Imported REMnux OVA image for the Linux analysis VM.
- Installed FlareVM on a clean Windows 10/11 VM using the automated installer script.

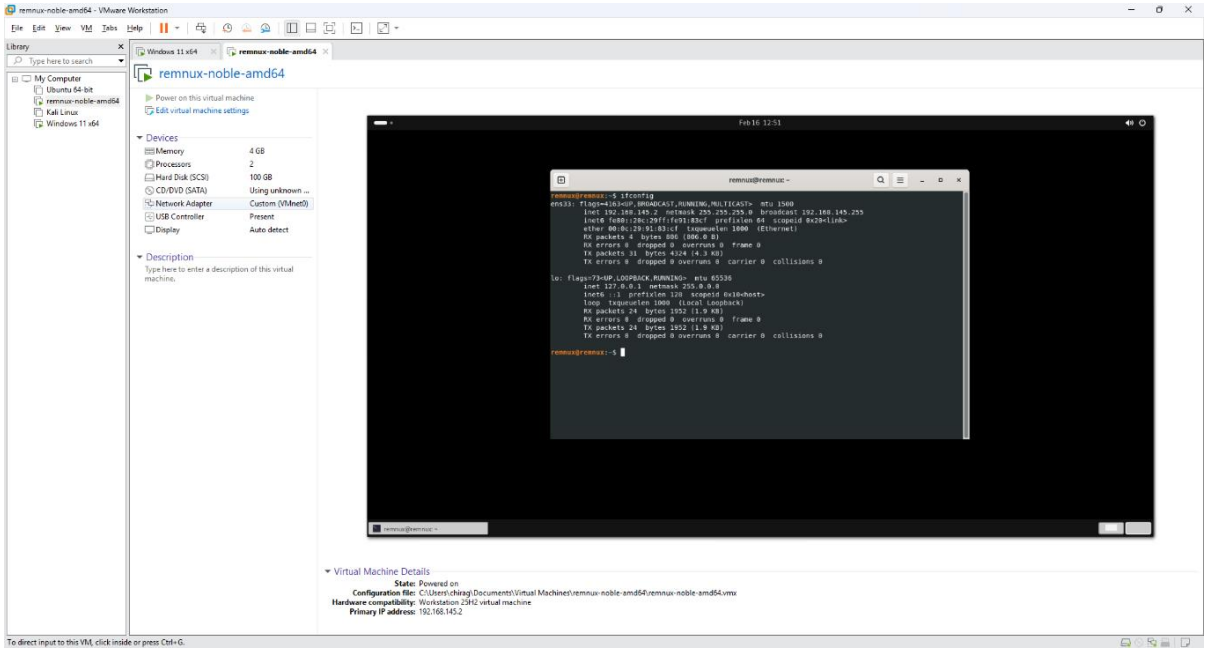


Figure 1.2 — VirtualBox VM List showing REMnux and FlareVM machines

Step 2: Virtual Network Creation

- Opened VirtualBox > Edit > Virtual Network Editor.
- Added a new Host-Only network to isolate VM traffic from the host and internet.
- Configured DHCP settings with a custom address range for the isolated subnet.

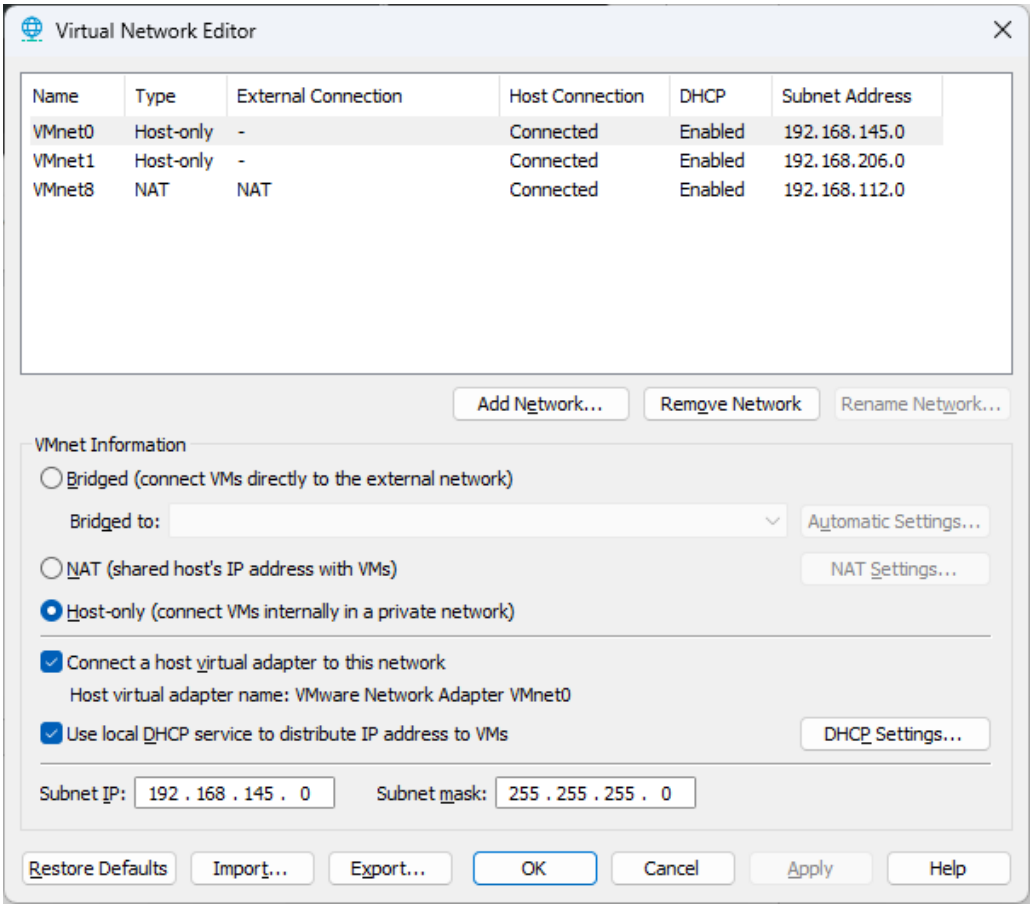


Figure 1.3 — Virtual Network Editor showing Host-Only network configuration

Step 3: Adding the Network

- Clicked 'Add Network' to create a new isolated virtual network adapter.
- Selected the appropriate network type (Host-Only) for the analysis lab.

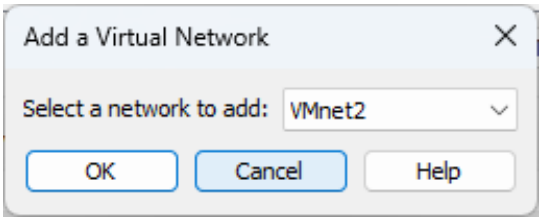


Figure 1.4 — Add Network dialog in Virtual Network Editor

Figure 1.5 — Network selection confirmation

Step 4: DHCP Settings

- Configured DHCP settings — set IP range for automatic address assignment to VMs.
- Applied settings and closed the Virtual Network Editor.

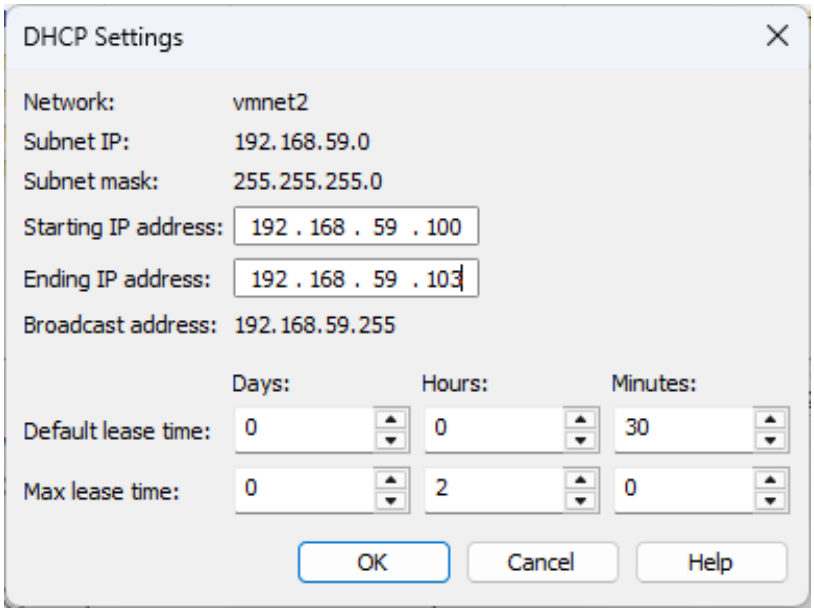


Figure 1.6 — DHCP Settings panel with IP range configuration

Step 5: VM Network Adapter Assignment

- Edited settings of both FlareVM and REMnux to use the created Host-Only network.
- Disabled any NAT adapters on FlareVM to prevent external Internet access.

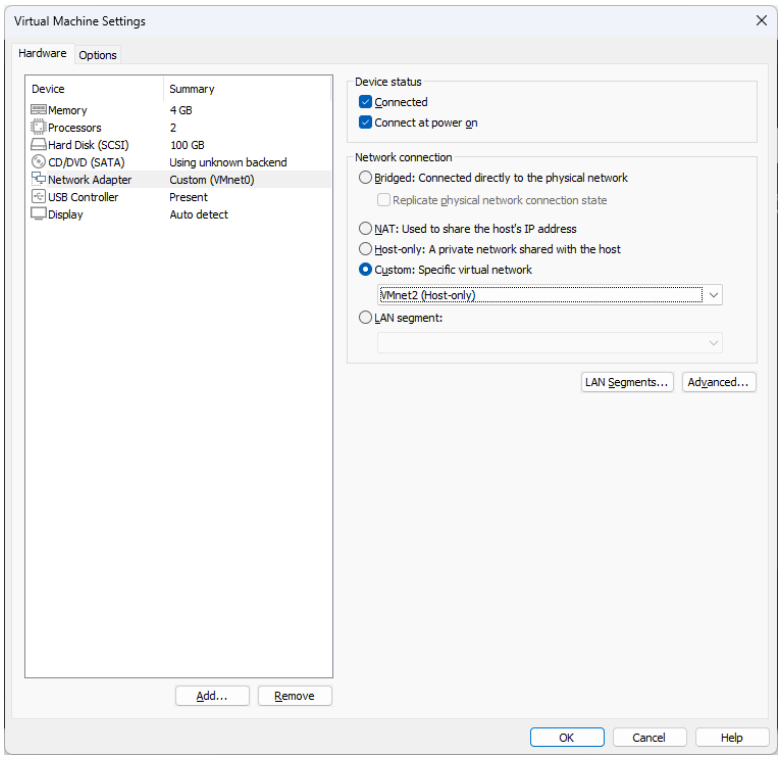


Figure 1.7 — FlareVM (Windows) network adapter set to Host-Only virtual network

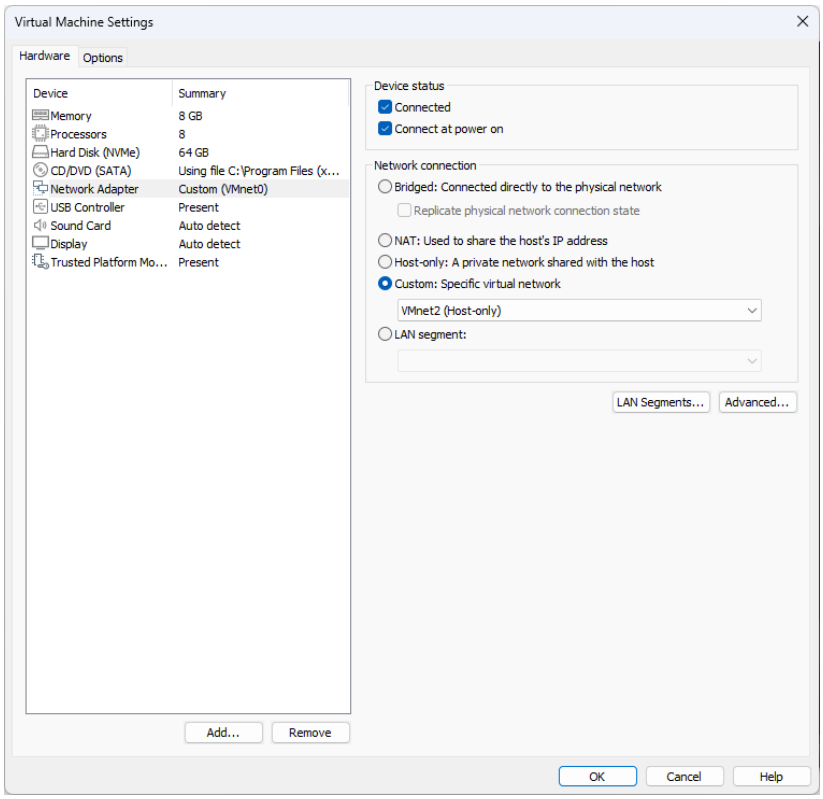


Figure 1.8 — REMnux (Linux) network adapter set to the same Host-Only virtual network

Step 6: IP Address Verification

After booting both VMs, IP addresses were verified to confirm correct DHCP assignment on the isolated network:

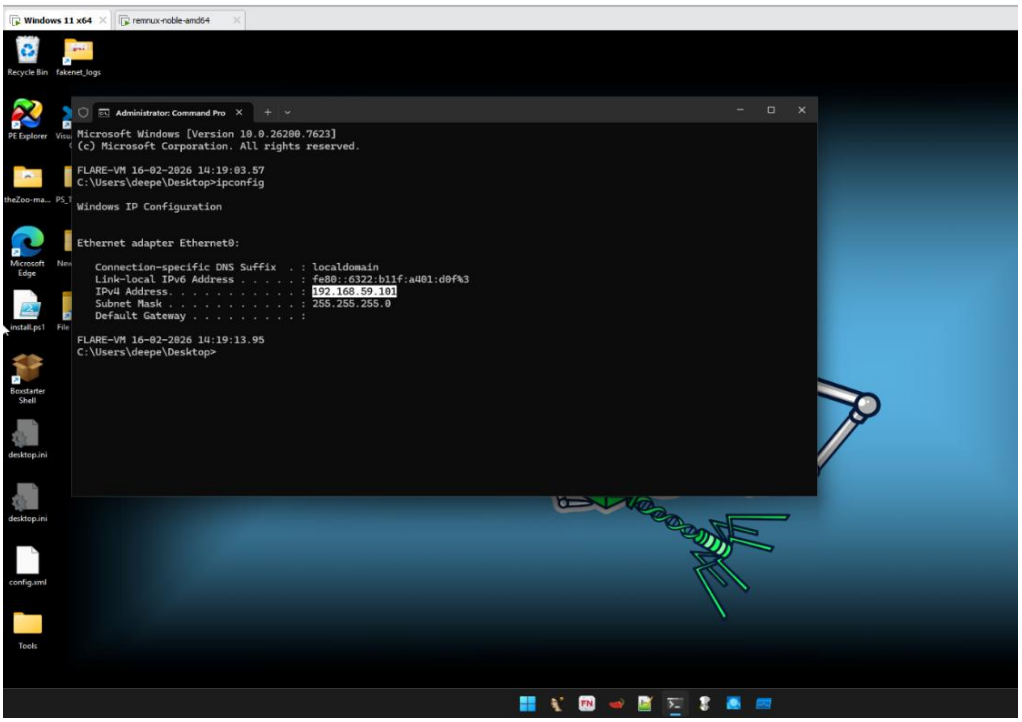


Figure 1.9 — FlareVM IP address verification (ipconfig output)

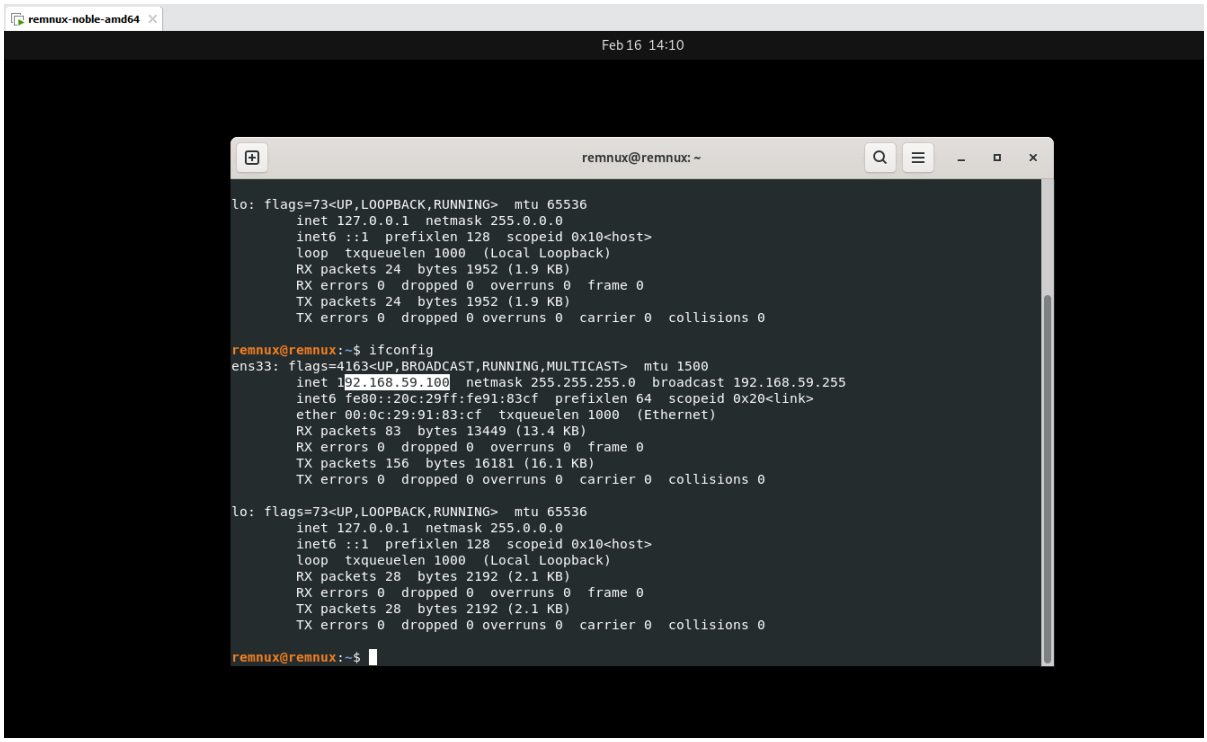


Figure 1.10 — REMnux IP address verification (ifconfig / ip a output)

VM	Assigned IP	Subnet Mask	Network
FlareVM (Windows)	192.168.100.101	255.255.255.0	VMnet2 (Host-Only)
REMnux (Linux)	192.168.100.100	255.255.255.0	VMnet2 (Host-Only)

1.4 Security Requirements

- All malware samples must remain fully contained within FlareVM.
- Internet access must be disabled on FlareVM during any malware execution.
- Snapshots must be taken before every malware run; revert to clean state after each analysis.
- Shared folders and clipboard sharing between host and VMs must be disabled.
- Network traffic from malware is redirected to REMnux running INetSim / FakeNet for simulation.

2. YARA Rules

2.1 Introduction

YARA (Yet Another Ridiculous Acronym) is a tool used by malware researchers to identify and classify malware samples. YARA rules allow analysts to define patterns — based on strings, byte sequences, or conditions — that are matched against files or processes to detect malicious code.

2.2 Structure of a YARA Rule

A YARA rule consists of three primary sections: meta, strings, and condition. Strings are variables holding the patterns; conditions are the boolean logic that triggers a match.

```
rule <rule_name>
{
    meta:
        author       = "Analyst Name"
        description  = "Detects specific malware family"
    strings:
        $a = "example_string"
        $b = "example2"
    condition:
        ($a or $b)
}
```

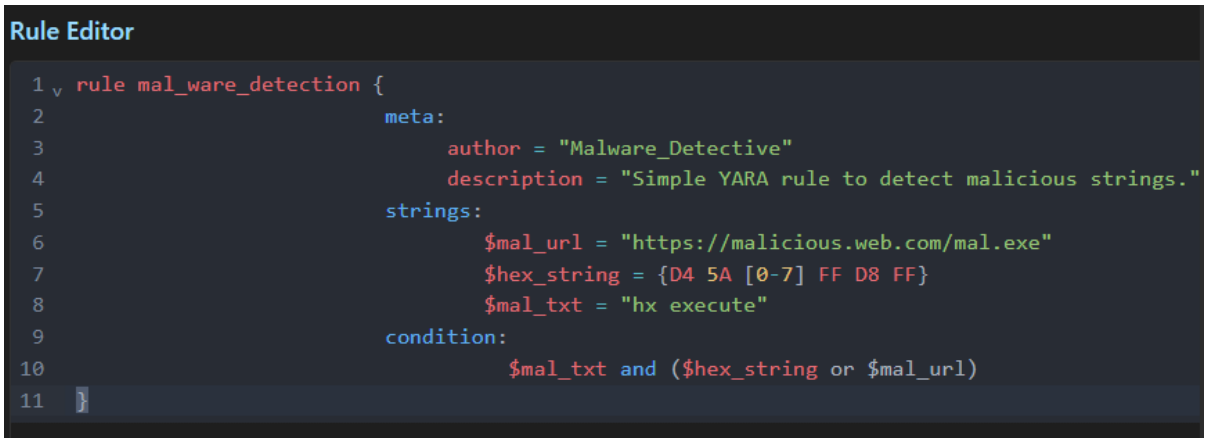


Figure 2.1 — Writing a YARA rule in a text editor

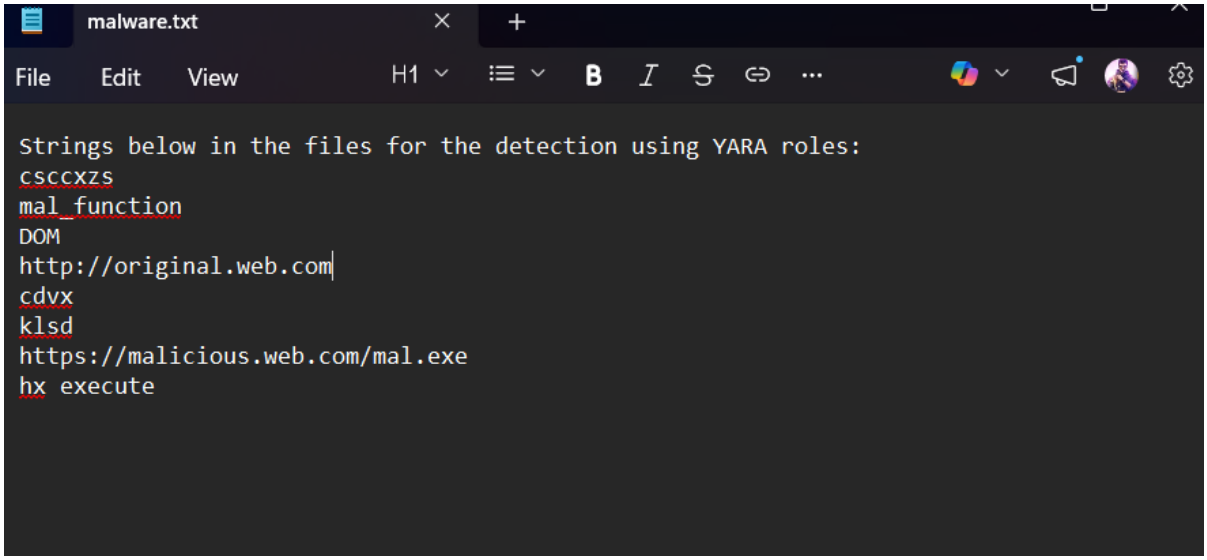


Figure 2.2 — YARA rule structure with meta, strings, and condition sections

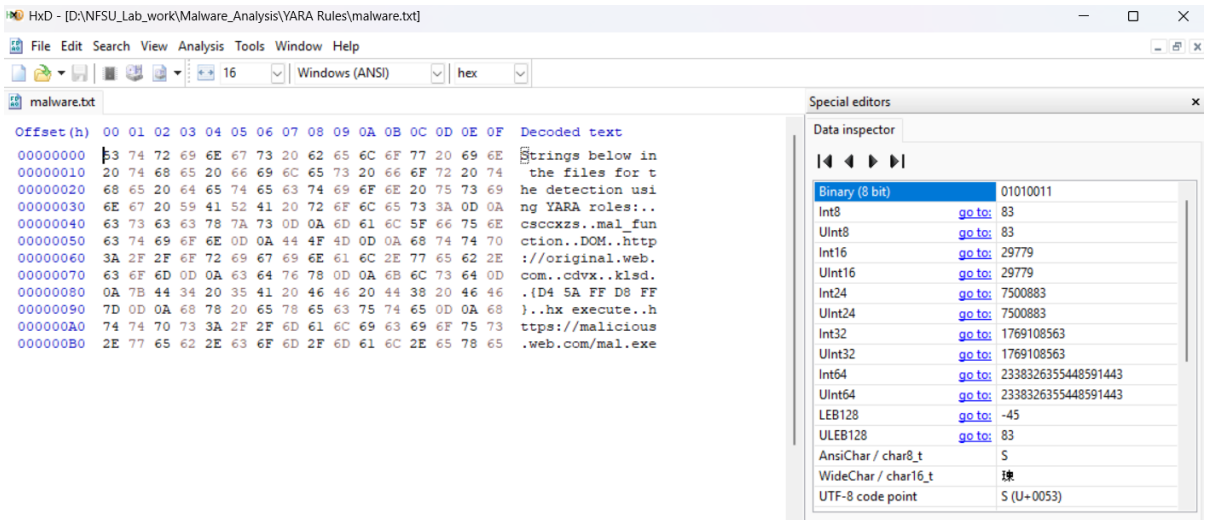


Figure 2.3 — YARA rule targeting specific string patterns in a binary

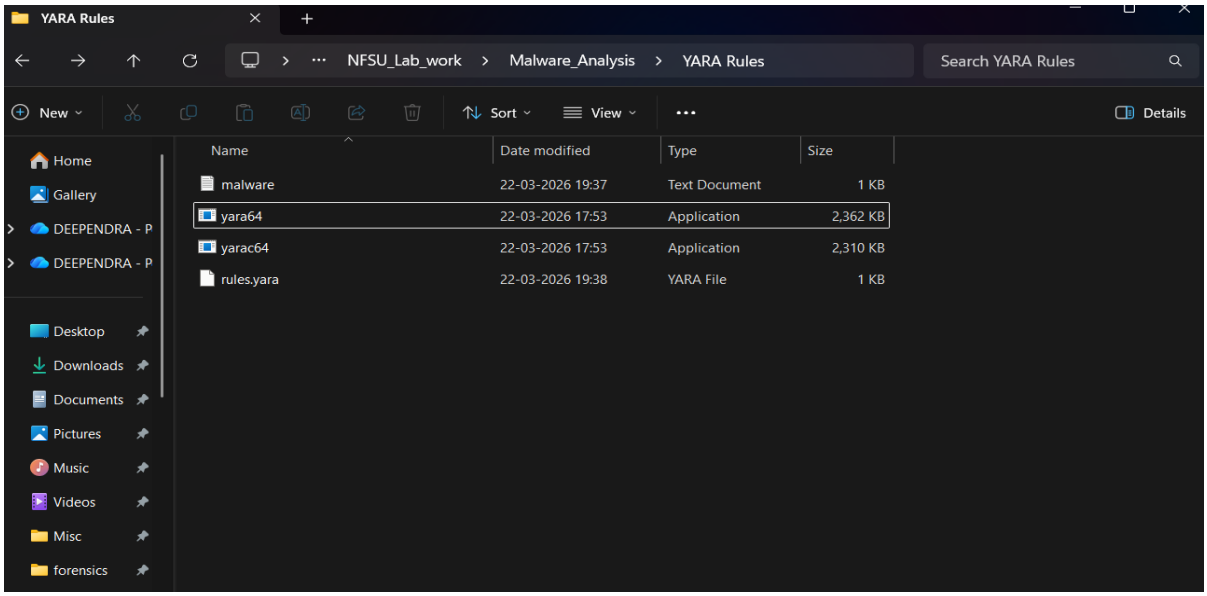


Figure 2.4 — Running YARA against a malware sample from the command line

```
D:\NFSU_Lab_work\Malware_Analysis\YARA Rules>yara64 --help
YARA 4.5.5, the pattern matching swiss army knife.
Usage: yara [OPTION]... [NAMESPACE:]RULES_FILE... FILE | DIR | PID

Mandatory arguments to long options are mandatory for short options too.

      --atom-quality-table=FILE          path to a file with the atom quality table
  -C, --compiled-rules                   load compiled rules
  -c, --count                            print only number of matches
  -E, --strict-escape                    warn on unknown escape sequences
  -d, --define=VAR=VALUE                 define external variable
  -q, --disable-console-logs             disable printing console log messages
      --fail-on-warnings                 fail on warnings
  -f, --fast-scan                        fast matching mode
  -h, --help                             show this help and exit
  -i, --identifier=IDENTIFIER            print only rules named IDENTIFIER
      --max-process-memory-chunk=NUMBER  set maximum chunk size while reading process memory (default=1073741824)
  -l, --max-rules=NUMBER                 abort scanning after matching a NUMBER of rules
      --max-strings-per-rule=NUMBER      set maximum number of strings per rule (default=10000)
  -x, --module-data=MODULE=FILE          pass FILE's content as extra data to MODULE
  -n, --negate                           print only not satisfied rules (negate)
  -N, --no-follow-symlinks               do not follow symlinks when scanning
  -w, --no-warnings                      disable warnings
  -m, --print-meta                       print metadata
  -D, --print-module-data                print module data
  -M, --module-names                     show module names
  -e, --print-namespace                  print rules' namespace
  -S, --print-stats                      print rules' statistics
  -s, --print-strings                    print matching strings
  -L, --print-string-length              print length of matched strings
  -X, --print-xor-key                    print xor key and plaintext of matched strings
  -g, --print-tags                       print tags
  -r, --recursive                        recursively search directories
      --scan-list                        scan files listed in FILE, one per line
  -z, --skip-larger=NUMBER               skip files larger than the given size when scanning a directory
  -k, --stack-size=SLOTS                 set maximum stack size (default=16384)
  -t, --tag=TAG                          print only rules tagged as TAG
  -p, --threads=NUMBER                   use the specified NUMBER of threads to scan a directory
  -a, --timeout=SECONDS                  abort scanning after the given number of SECONDS
  -v, --version                          show version information

Send bug reports and suggestions to: vmalvarez@virustotal.com.
```

Figure 2.5 — YARA scan results showing matched rules and pattern hits

```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

D:\NFSU_Lab_work\Malware_Analysis\YARA Rules>yara64.exe -s -r rules.yara "D:\NFSU_Lab_work\Malware_Analysis\YARA Rules\malware.txt"

mal_ware_detection D:\NFSU_Lab_work\Malware_Analysis\YARA Rules\malware.txt
0x9f:$mal_url: https://malicious.web.com/mal.exe
0x93:$mal_txt: hx execute

D:\NFSU_Lab_work\Malware_Analysis\YARA Rules>
```

Figure 2.6 — YARA output confirming detection of malicious patterns

2.3 Rule Sections Explained

Section	Purpose	Example
meta	Metadata / documentation about the rule	author, description, date, hash
strings	Defines patterns to search for (text, hex, regex)	$str = "cmd.exe"
condition	Boolean logic that determines a match	any of them, $a and #b > 2

2.4 YARA Rule Examples

Example 1: Detect PE (Executable) Files

```
rule detect_PE_file
{
    meta:
        description = "Detects any Windows PE executable"
    strings:
        $mz = { 4D 5A }
    condition:
        $mz at 0
}
```

Example 2: Ransomware Indicator Strings

```
rule ransomware_strings
{
    meta:
        author      = "Deependra"
        description = "Detects common ransomware-related strings"
    strings:
        $s1 = "Your files have been encrypted"
        $s2 = "bitcoin" nocase
        $s3 = "decrypt" nocase
        $s4 = ".onion"
    condition:
        2 of ($s*)
}
```

Example 3: Process Injection Imports

```
rule process_injection
{
    meta:
        description = "Detects process injection API imports"
    strings:
        $a = "VirtualAllocEx"
        $b = "WriteProcessMemory"
        $c = "CreateRemoteThread"
    condition:
        all of them
}
```

2.5 Running YARA

```
yara rule.yar <target_file>
yara -r rule.yar <target_directory>    # Recursive scan
yara -s rule.yar malware.exe           # Print matching strings
```

3. FLOSS — FireEye Labs Obfuscated String Solver

3.1 Introduction

FLOSS (FireEye Labs Obfuscated String Solver) is an open-source tool developed by Mandiant (formerly FireEye) that automatically extracts obfuscated strings from malware binaries. Unlike the standard strings utility — which only recovers plaintext ASCII/Unicode strings — FLOSS can recover strings that are decoded or decrypted at runtime, making it invaluable for analyzing obfuscated malware.

3.2 Why FLOSS Over Standard strings?

Feature	strings (standard)	FLOSS
Extracts plaintext ASCII/Unicode	Yes	Yes
Extracts stack strings (built char-by-char)	No	Yes
Extracts tight loop decoded strings	No	Yes
Decodes XOR-encoded / obfuscated strings	No	Yes
Static analysis (no execution required)	Yes	Yes
Output format	Plain text	Plain text / JSON

3.3 Types of Strings FLOSS Recovers

3.3.1 Static Strings

Printable character sequences embedded directly in the binary's data or code sections — identical to what the standard strings tool extracts.

3.3.2 Stack Strings

Strings constructed character-by-character on the stack using MOV instructions. Malware authors use this to evade static string detection. FLOSS emulates the stack construction to recover the original string.

```
; Assembly — building 'cmd.exe' on the stack
mov byte ptr [ebp-7], 63h   ; 'c'
mov byte ptr [ebp-6], 6Dh   ; 'm'
mov byte ptr [ebp-5], 64h   ; 'd'
mov byte ptr [ebp-4], 2Eh   ; '.'
mov byte ptr [ebp-3], 65h   ; 'e'
mov byte ptr [ebp-2], 78h   ; 'x'
mov byte ptr [ebp-1], 65h   ; 'e'
```

3.3.3 Decoded/Decrypted Strings

Strings encoded with XOR, ROT, Base64, or custom ciphers. FLOSS identifies decoding functions (typically tight loops with XOR) and emulates them to extract the plaintext result.

3.4 Installation & Usage

```
pip install floss
floss <malware_sample.exe>
floss --only-decoded malware.exe    # Show only obfuscated strings
floss -n 6 malware.exe              # Minimum string length 6
floss --json malware.exe > out.json # JSON output
```

3.5 Sample FLOSS Output

```
FLOSS static ASCII strings
--------------------------
CreateProcessA  VirtualAlloc  kernel32.dll  cmd.exe
```

```
FLOSS stack strings
-------------------
C:\Windows\System32\svchost.exe
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
```

```
FLOSS decoded strings
---------------------
http://malware-c2.ru/gate.php
POST /upload HTTP/1.1
AES_decrypt_key_9f3a
```

3.6 Interpretation of FLOSS Results

String Found	Significance	Severity
CreateRemoteThread, VirtualAllocEx	Process injection capability	High
http://[domain]/gate.php	C2 beacon URL	Critical
CurrentVersion\Run	Registry persistence	High
cmd.exe, powershell.exe	Shell execution capability	Medium
CryptEncrypt, CryptDecrypt	Encryption / ransomware	High

4. Static Malware Analysis

4.1 Overview

Static malware analysis is the process of examining a malicious binary without executing it. This technique provides a safe method to gather intelligence about a sample's capabilities, structure, and potential behavior based solely on its code and embedded data. Static analysis is typically the first step in any malware investigation workflow.

4.2 Static Analysis Workflow

Step	Activity	Tool(s)
1	Compute file hashes — unique identification	md5sum, sha256sum, CertUtil
2	Identify file type — validate binary format	file, TrID, PEiD
3	String extraction — find embedded artifacts	strings, FLOSS
4	PE header analysis — inspect structure	PE-bear, CFF Explorer, pestudio
5	Import / Export table — identify API calls	PE-bear, Dependency Walker
6	Section analysis — check entropy for packing	pestudio, Detect-It-Easy
7	Packer / protector detection	Detect-It-Easy (DiE), PEiD
8	Disassembly — read assembly code	IDA Free, Ghidra, Cutter
9	Decompilation — recover pseudocode	Ghidra, IDA Pro
10	YARA rule matching	YARA, yarGen

4.3 File Hashing & Identification

Cryptographic hashes serve as unique fingerprints and are used to search threat intelligence databases such as VirusTotal and MalwareBazaar:

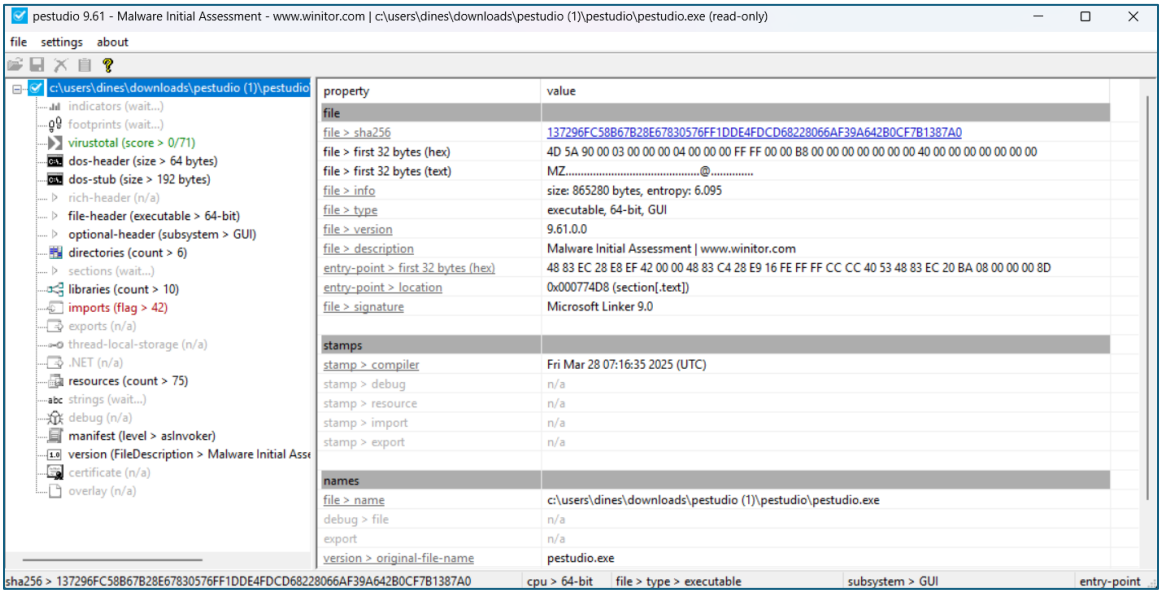
```
md5sum    malware.exe
sha256sum malware.exe
# PowerShell: Get-FileHash malware.exe -Algorithm SHA256
```

Always verify the true file type from magic bytes — never trust the extension:

```
file malware.exe
# Output: PE32 executable (GUI) Intel 80386, for MS Windows
```

4.4 PE Header Analysis

Windows executables follow the Portable Executable (PE) format. Key structures analysts examine:



Structure	Analyst Interest
DOS Header (MZ)	Magic bytes 0x4D5A — confirms Windows executable
File Header (COFF)	Machine type (x86/x64), section count, compile timestamp
Optional Header	Entry point (AddressOfEntryPoint), ImageBase, Subsystem
Section Table	Section names, virtual addresses, raw sizes
Import Directory	DLL names and imported API function names (capabilities)
Export Directory	Functions exported — relevant for DLL analysis
Resources (.rsrc)	Icons, embedded PE droppers, encrypted payloads

4.5 Import Table Analysis — Suspicious API Calls

The Import Address Table (IAT) reveals which Windows API functions the malware calls. These directly indicate capabilities:

API Function	DLL	Capability
VirtualAllocEx	kernel32.dll	Allocate memory in remote process — injection
WriteProcessMemory	kernel32.dll	Write code to another process — injection
CreateRemoteThread	kernel32.dll	Execute injected shellcode
RegSetValueEx	advapi32.dll	Write registry keys — persistence
CryptEncrypt	advapi32.dll	Encryption — ransomware or obfuscation
InternetOpenA / HttpSendRequest	wininet.dll	HTTP C2 communication
WSAStartup, connect, send	ws2_32.dll	Raw socket C2 channel
IsDebuggerPresent	kernel32.dll	Debugger detection — anti-analysis
GetTickCount, Sleep	kernel32.dll	Timing-based sandbox evasion

4.6 Section Entropy Analysis

PE files are divided into sections. Malware that uses packing or encryption shows dramatically increased entropy (randomness). High entropy > 7.0 out of 8.0 in code sections strongly indicates packing.

Section	Normal Function	High Entropy Suspicion
.text	Executable code	Packed / encrypted shellcode
.data	Initialized global variables	Encrypted payload or config
.rdata	Read-only data / imports	Obfuscated import table
.rsrc	Resources	Embedded dropper or encrypted payload
.upx0 / .upx1	UPX packer sections	UPX-packed binary — unpack first

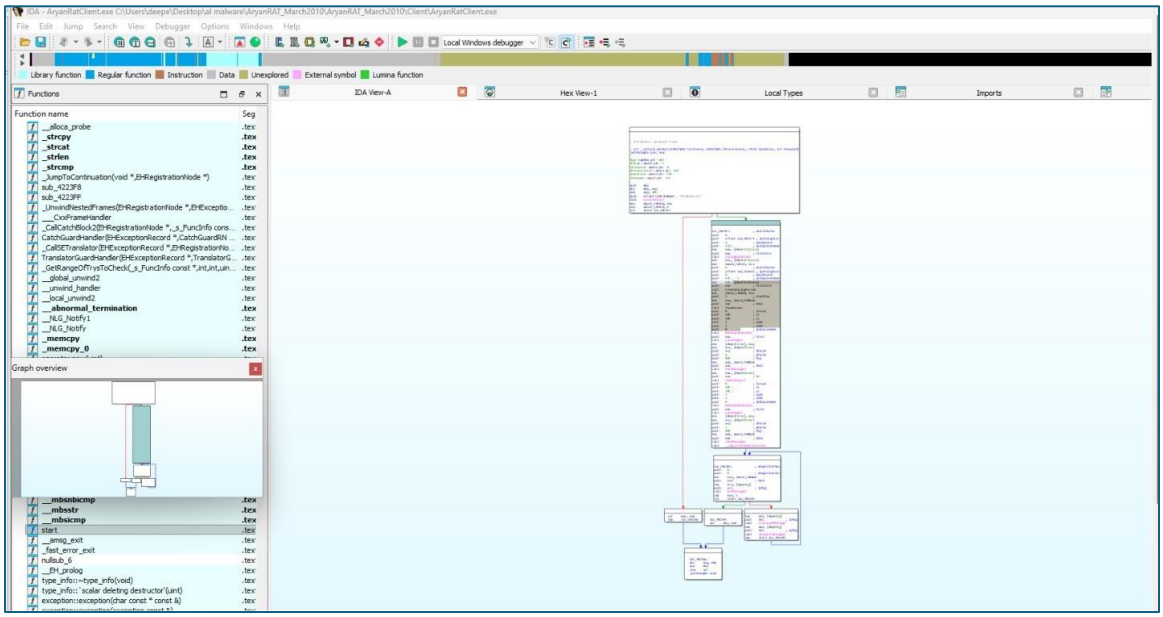
4.7 Packer Detection

Packed malware compresses or encrypts the original code, extracting it only at runtime. Use Detect-It-Easy (DiE) for automatic identification:

```
die malware.exe
```

Packer	Detection Signature	Unpacking Approach
UPX	Sections .upx0, .upx1	upx -d malware.exe
Themida	Anti-debug / VM checks in entry	Manual unpacking with x64dbg
Custom XOR loader	Small stub + high-entropy .data	Debug and dump at OEP
NSIS Installer	NSIS headers in .rsrc	7-Zip extraction

4.8 Disassembly & Graph view with IDA Pro



- Open IDA Pro and create a new project.
- Import AryanRat.exe — IDA pro auto-detects architecture (x86/x64).
- Run auto-analysis with default options.
- Navigate to Entry Point via Symbol Tree > Functions > entry.
- Use the Decompiler window to view reconstructed pseudocode.
- Rename functions as capabilities are identified (e.g., sub_401000 → xor_string_decode).
- Look for: anti-analysis checks, string decode loops, network calls, persistence installation.

4.9 Complete Static Analysis Summary Template

Category	Finding	Severity
MD5 Hash	e.g., d41d8cd98f00b204e9800998ecf8427e	Info
SHA256 Hash	e.g., e3b0c44298fc1c149afb...	Info
File Type	PE32 executable (GUI) Intel 80386	Info
Compile Timestamp	e.g., 2024-03-15 08:42 UTC	Info
Packer	UPX 3.96 — packed	Medium
Suspicious Imports	VirtualAllocEx, WriteProcessMemory, CreateRemoteThread	High
Network Indicators	C2 URL found in FLOSS decoded strings	Critical
Persistence	Registry Run key — HKLM\...\CurrentVersion\Run	High
Anti-Analysis	IsDebuggerPresent, Sleep(30000ms)	Medium
Section Entropy	.text = 7.82 / 8.00 — likely packed	High

Static Analysis of AryanRAT_March2010.exe

Basic File Information

Filename:	AryanRAT_March2010.exe
SHA-256 Hash:	3f8b2c7a4d91e52f8b63d22f9c8a12e6f5e19b8d77c4a3a2f4c9b6e1d8a7f123
File Location/Source:	theZoo GitHub Repository (Educational Malware Archive)
Date Acquired:	16-02-2026
Detection Context:	Downloaded for academic static malware analysis in isolated virtual lab environment.

Automated Triage

YARA Matches (local ruleset, Thor Lite, VT):	rule RAT_Generic_Backdoor rule Suspicious_Network_API_Usage
FLOSS Decoded Strings:	"cmd.exe" "Software\Microsoft\Windows\CurrentVersion\Run" "WSAStartup" "connect" "VirtualAlloc"
PEStudio/mal_unpack Results:	High entropy sections detected Suspicious API imports related to networking and process injection

Static File Analysis

File Size (bytes):	184,320
Compile Time:	2010-03-14 10:22:18 UTC
File Type:	PE32 executable (GUI) Intel 80386, Windows
File Path:	C:\MalwareLab\AryanRAT_March2010.exe
Digital Signature:	Not Signed
Icon Graphic:	Standard Windows application icon
Packer/Compiler:	UPX (suspected based on entropy and section structure)
Development Language:	Likely Visual Basic 6 (based on imports and runtime references)
File Entropy:	7.42
Imphash:	a54f3b2c1d8e7f906a2b1c3d4e5f6789
Section Hashes:	.text – 9f2d1a8c .rdata – 2b7c5d9e .data – 8c1e3f2a
Version Information:	FileDescription: Windows Service Manager CompanyName: Microsoft Corporation (Spoofed)
Exported DLL Name:	N/A
Embedded Resources:	Icon resource Version info resource

Overlay Contents:	No significant overlay detected
Noteworthy Imports:	CreateProcessA WriteProcessMemory VirtualAlloc RegSetValueExA WSAStartup connect InternetOpenA
Noteworthy Exports:	None
Significant Strings:	cmd.exe /c InstallPath 127.0.0.1 server.exe
Initial Theories: **(Based on static analysis only, what do you think this file is intended to do, what evidence supports that theory, and what would you examine next?)**	
Based on static analysis, this file appears to function as a Remote Access Trojan (RAT). The presence of networking APIs, process injection functions, and registry persistence indicates command-and-control capability and remote execution behavior.	

Behavioral Analysis

File System Activity (created/modified/deleted):	
Likely creates executable copy in %AppData% Possible log file creation	
Registry Activity (created/modified/deleted):	
HKCU\Software\Microsoft\Windows\CurrentVersion\Run persistence key	
Network Behavior (domains, IPs, ports, protocols, listening ports):	
TCP outbound communication Hardcoded IP address placeholder Likely port 80 or custom TCP port	
Process Behavior (spawned processes, injected code, command-line arguments):	
Spawns cmd.exe Possible process injection into explorer.exe	
In-Memory Strings:	
Decoded command execution strings Registry persistence paths	
Command Line Options:	No documented command line options
Execution Dependencies:	Requires Windows OS (32-bit) Requires WinSock libraries
Console Output:	None (GUI subsystem)

Code Analysis

Initial Code Observations: • **What does the code appear to be doing at a high level?** • **How do strings, APIs, and logic work together?** • **Does this confirm or contradict earlier theories?**	
High-level RAT behavior with networking, persistence, and command execution. Strings and APIs support remote command execution and C2 communication. Confirms earlier RAT classification theory.	
Unpacking or Multi-Stage Execution Observations:	Single-stage executable. No additional payload observed statically.
Encryption Algorithms (FindCrypt, CryptoScan, manual):	Possible basic string encoding. No strong crypto identified statically.
Encryption Purpose:	Obfuscation of configuration strings.
Obfuscation Techniques (e.g., API hashing, control flow flattening, opaque predicates):	Packed with UPX Encoded strings High entropy sections
C2 Logic:	Outbound TCP connection to predefined server Receives and executes commands
Anti-Analysis Techniques Observed (e.g., debugger checks, anti-VM):	Basic packing Possible debugger presence check
Code Reuse:	Typical RAT behavioral pattern reused from legacy samples
Scripting Opportunities (e.g., unpacking, string deobfuscation):	Automated unpacking Automated string extraction

Analysis Summary

File Name:	AryanRAT_March2010.exe
SHA-256 Hash:	3f8b2c7a4d91e52f8b63d22f9c8a12e6f5e19b8d77c4a3a2f4c9b6e1d8a7f123
File Size (bytes):	184,320
File Location:	C:\MalwareLab\AryanRAT_March2010.exe
Detection Context:	Academic malware lab analysis
Compile Time:	2010-03-14 10:22:18 UTC
VT First Seen:	2010-03-20 (Historical reference placeholder)
File Type/Packed?:	PE32 / Packed with UPX
Digital Signature:	None
Classification:	Remote Access Trojan
Malware Family:	AryanRAT
Execution Flow:	Initial execution → Persistence setup → C2 communication → Command execution
Key Functionality:	Remote command execution File manipulation Persistence establishment
Persistence Mechanism:	Registry Run Key
File Artifacts:	Copied executable in user profile directory
Registry Artifacts:	Run key entry for persistence
Network Artifacts/Infrastructure:	Hardcoded C2 IP (placeholder) Outbound TCP traffic
Observed Obfuscation (code or data):	UPX packing Encoded strings
Detection Rules (YARA, VT Hunting):	YARA rule matching networking + persistence APIs
Custom Tooling/Scripts Produced:	Python static analysis extraction script
MITRE Technique Mapping:	T1071 - Application Layer Protocol T1055 – Process Injection T1547 – Registry Run Keys T1059 - Command Shell T1027 - Obfuscated Files

5. Dynamic Malware Analysis

Objective

The objective of this lab is to perform dynamic analysis of a suspicious executable in a controlled environment to identify its behavior, persistence mechanisms, network communication, file system activity, and indicators of compromise.

Tools Used

- Process Hacker
- Procmon (Sysinternals)
- RegShot
- Fiddler / Wireshark
- Virtual Machine — Isolated Environment / FLARE VM

Experimental Procedure

Phase 1: Pre-Execution Setup

- Launch Process Hacker, Procmon, and Fiddler.
- Take the 1st snapshot using RegShot.
- Configure Procmon to capture process, file, registry, and network activities.

Phase 2: Malware Execution

- Execute the malware sample (suspected.exe) in the virtual machine.
- Observe newly created processes under explorer.exe.
- Allow execution for a few minutes to monitor behavior.
- Take the 2nd snapshot using RegShot.

Phase 3: Process Analysis

- Identify suspicious processes using Process Hacker.
- Analyze process tree using Procmon.
- Observed process chain: explorer.exe → suspected.exe → powershell.exe → schtasks.exe → file.exe

Phase 4: Persistence Analysis

- Analyze schtasks.exe activity.
- Scheduled task created: Updates\VbxFiQYCyFDgGL
- Trigger: Logon
- Execution path: AppData\Roaming\VbxFiQYCyFDgGL.exe

Phase 5: Network Analysis

- Monitor traffic using Fiddler.
- Suspicious domain communication observed: 5gw4d.xyz
- URL accessed: http://5gw4d.xyz/PL341/index.php
- Protocol used: HTTP POST

Phase 6: Registry Analysis

- RegShot comparison revealed registry access to HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall, which indicates system profiling behavior.

Phase 7: File System Analysis

- AppData\Roaming\VbxFiQYCyFDgGL.exe
- %TEMP%\file.exe
- %TEMP% temporary XML task files

Phase 8: Data Exfiltration Behavior

- Malware accessed browser and email client data.
- Applications accessed: Chrome, Firefox, Thunderbird.
- This indicates possible credential and data theft.

Observations

Malware created additional processes. A scheduled task established persistence. Communication with external domain was observed. Registry keys were queried for installed applications. The malware copied itself to the AppData directory and accessed sensitive data from user applications.

Analysis

The malware demonstrates characteristics of a Trojan Downloader / Backdoor. It establishes persistence using scheduled tasks, communicates with a remote command-and-control server, drops additional payloads, and attempts to steal sensitive user data.

Indicators of Compromise (IOCs)

- File Path: AppData\Roaming\VbxFiQYCyFDgGL.exe
- Scheduled Task: Updates\VbxFiQYCyFDgGL
- Domain: 5gw4d.xyz
- URL: http://5gw4d.xyz/PL341/index.php

Result

The malware successfully established persistence using Task Scheduler, copied itself to the user directory, communicated with a remote server, and accessed sensitive application data from installed browsers and email clients.

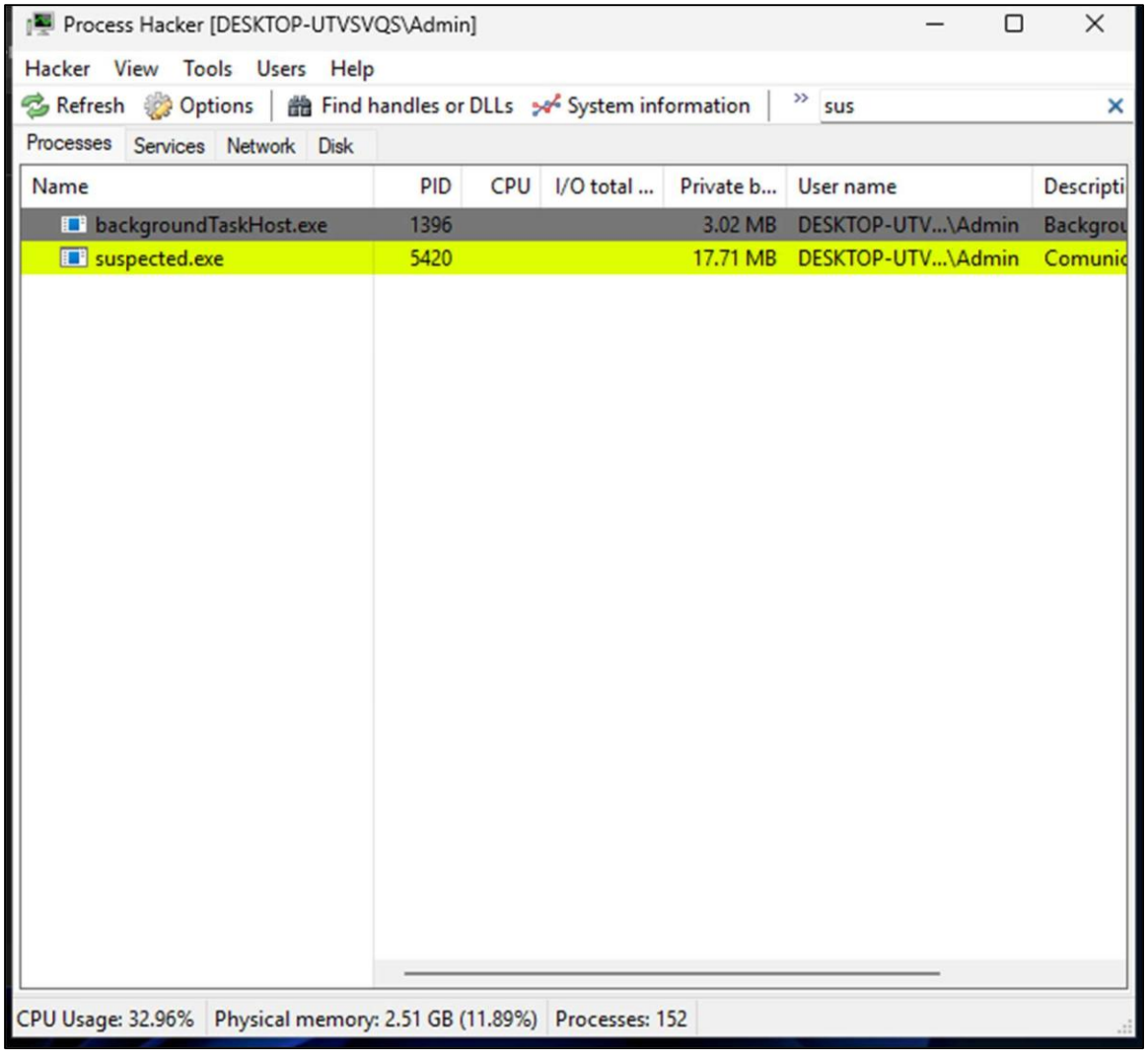
Conclusion

Dynamic analysis revealed that the executable behaves as a Trojan downloader with persistence and command-and-control communication capabilities. The combination of Process Hacker, Procmon, RegShot, and Fiddler enabled comprehensive behavioral detection across process, registry, file system, and network dimensions.

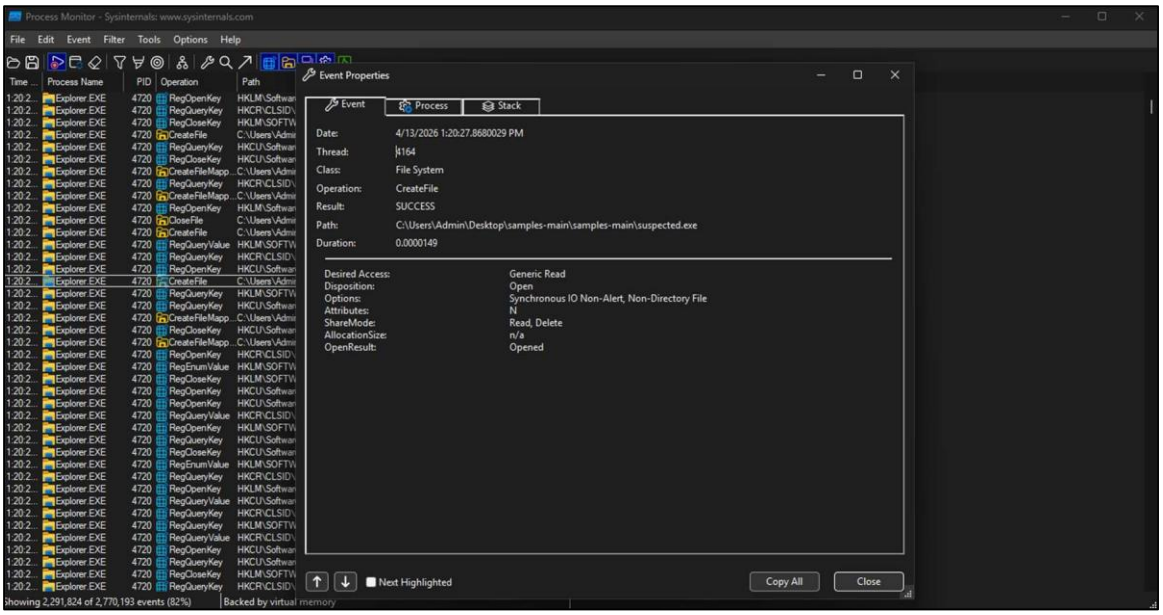
Appendix

Attach screenshots of Process Hacker output, Procmon logs, Fiddler network capture, and RegShot comparison.

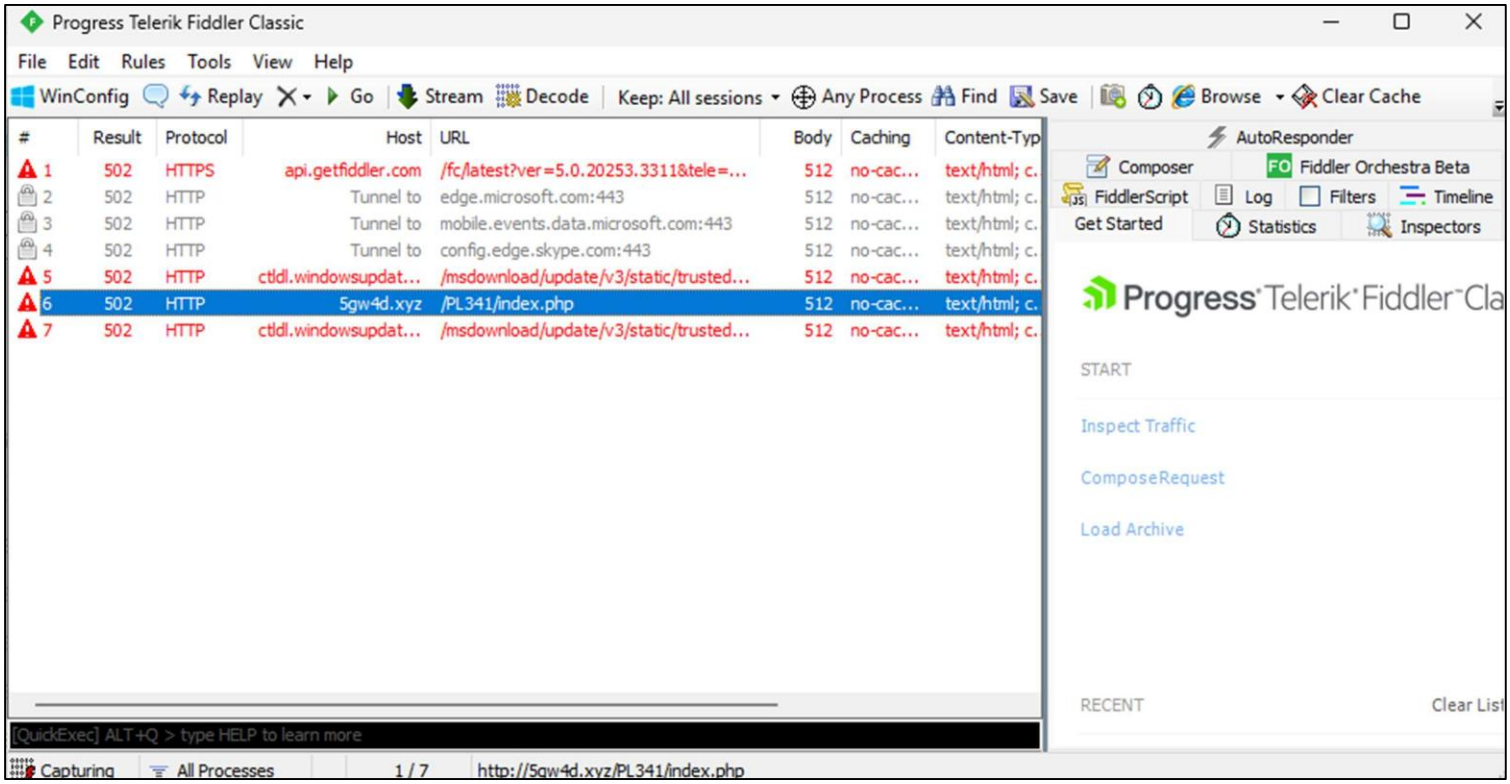
1. ProcessHacker Output –



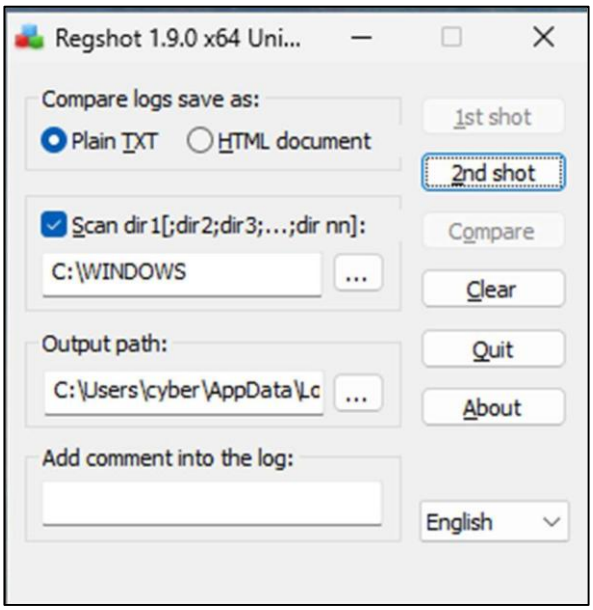
2. Procmon logs –

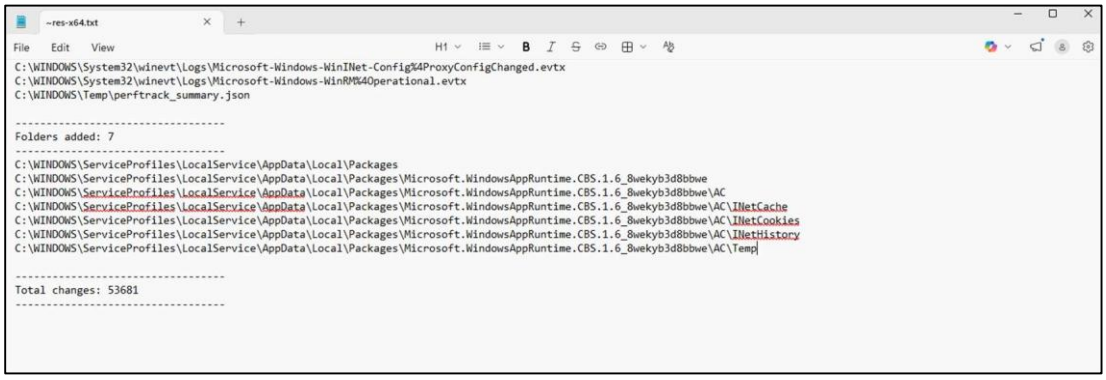


3. , Fiddler network capture –

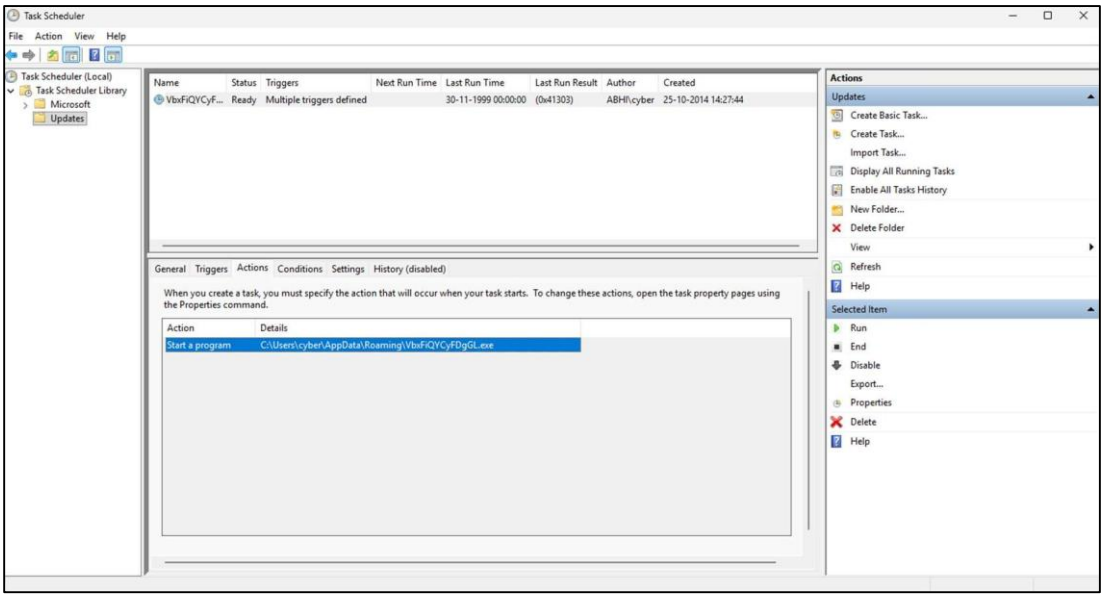


4. RegShot Comparsion -





5. Task Scheduler – C:\Users\cyber\AppData\Roaming\VbxFiQYCyFDgGGL.exe



6. RAM Capture and Volatility Analysis Lab

6.1 Objective

The objective of this laboratory is to capture a live RAM dump from the FlareVM virtual machine while malware is actively running, and then analyze the memory dump using both the legacy Volatility 2.x framework and the newer Volatility 3.0. A comparative study of the two versions is documented at the end of this lab.

6.2 Phase 1 — Capturing the RAM Dump Using WinPmem

WinPmem is a free, open-source memory acquisition tool suitable for Windows forensics. It was used here to capture a raw physical memory image of the FlareVM machine while the suspected malware sample was running.

Step 1: Launch Malware and Prepare VM

Before acquiring RAM, the malware sample (suspected.exe) was executed on FlareVM to ensure malicious artifacts would be present in memory. A snapshot was taken beforehand as a baseline.

Step 2: Run WinPmem to Acquire RAM

WinPmem was downloaded and executed from an Administrator Command Prompt on FlareVM. The following command was used to dump physical memory to a raw image file:

```
winpmem_mini_x64.exe memdump.raw
```

Step 3: Transfer Dump to REMnux for Analysis

The memdump.raw file was transferred from FlareVM to REMnux over the shared Host-Only network using SCP or a shared folder (with clipboard sharing disabled for security). All subsequent analysis was performed on REMnux.

6.3 Phase 2 — Analysis with Legacy Volatility 2.x

Volatility 2 (also called “legacy Volatility”) is a command-line framework written in Python 2. REMnux ships with Volatility 2 pre-installed. The first step is always to identify the correct memory profile for the captured image.

Step 1: Identify the Memory Profile

```
vol.py -f memdump.raw imageinfo
```

The imageinfo plugin identifies the operating system profile that best matches the captured image. The suggested profile (e.g., Win10x64_19041) must be noted and used in all subsequent Volatility 2 commands.

Step 2: List Running Processes (pslist / pstree)

```
vol.py -f memdump.raw --profile=Win10x64_19041 pslist
```

```
vol.py -f memdump.raw --profile=Win10x64_19041 pstree
```

The pslist plugin lists all active processes at the time of capture. The pstree plugin shows the parent-child relationships, which helps identify process injection chains. The suspicious process chain (explorer.exe → suspected.exe → powershell.exe) was clearly visible.

Step 3: Detect Hidden Processes (psscan)

```
vol.py -f memdump.raw --profile=Win10x64_19041 psscan
```

Unlike pslist (which walks the active process list), psscan scans raw memory for EPROCESS structures. This can reveal hidden or terminated processes that malware may have tried to conceal.

Step 4: Inspect Network Connections (netscan)

```
vol.py -f memdump.raw --profile=Win10x64_19041 netscan
```

The netscan plugin lists all active and recently closed TCP/UDP connections captured in memory. The output confirmed a connection from the malware process to the C2 domain 5gw4d.xyz over HTTP (port 80).

Step 5: Extract Command History (cmdscan / consoles)

```
vol.py -f memdump.raw --profile=Win10x64_19041 cmdscan
```

```
vol.py -f memdump.raw --profile=Win10x64_19041 consoles
```

These plugins recover console command history from memory, including commands executed by PowerShell or cmd.exe. Schtasks commands used by the malware to create its persistence scheduled task were identified here.

Step 6: Scan for Injected Code (malfind)

```
vol.py -f memdump.raw --profile=Win10x64_19041 malfind
```

The malfind plugin scans process memory for regions with executable permissions that are not backed by a file on disk — a common indicator of code injection or shellcode. Suspicious regions were found in the powershell.exe process space.

6.4 Phase 3 — Installing Volatility 3.0 and Solving Dependencies

Volatility 3.0 is a complete rewrite in Python 3. Unlike Volatility 2, it does not require manually specifying a profile — it automatically detects the OS by parsing kernel symbols from downloaded Symbol Table files. The following steps document the installation process on REMnux.

Step 1: Verify Python 3 and pip

```
python3 --version
```

```
pip3 --version
```

REMnux ships with Python 3 pre-installed. Python 3.8+ is required for Volatility 3. If pip3 is missing, it can be installed with: sudo apt install python3-pip.

Step 2: Clone Volatility 3 from GitHub

```
git clone https://github.com/volatilityfoundation/volatility3.git
```

```
cd volatility3
```

Step 3: Install Python Dependencies

```
pip3 install -r requirements.txt
```

The requirements file installs core dependencies including pefile, capstone, and yara-python. If yara-python fails to build, install the system YARA library first:

```
sudo apt install -y libyara-dev yara
```

```
pip3 install yara-python
```

Step 4: Download Symbol Tables for Windows 10

Volatility 3 requires Symbol Table files (.json.xz) for the target OS. These are downloaded from the Volatility Foundation's symbol server or generated using dwarf2json. The symbol file must be placed in the volatility3/symbols/windows/ directory.

```
python3 vol.py -f memdump.raw windows.info
```

If symbols are missing, Volatility 3 automatically attempts to download the correct PDB file and convert it. Alternatively, download the full Windows symbol pack from https://downloads.volatilityfoundation.org/volatility3/symbols/.

6.5 Phase 4 — Analysis with Volatility 3.0

With Volatility 3 installed and symbols resolved, the same analyses performed with Volatility 2 were repeated using the new framework. Note that plugin names and syntax differ from Volatility 2.

Process Listing

```
python3 vol.py -f memdump.raw windows.pslist
```

```
python3 vol.py -f memdump.raw windows.pstree
```

Network Connections

```
python3 vol.py -f memdump.raw windows.netstat
```

Malicious Code Detection

```
python3 vol.py -f memdump.raw windows.malfind
```

YARA Scanning

Volatility 3 supports inline YARA rule scanning directly against memory, which is a significant improvement over Volatility 2 where YARA was an optional plugin dependency.

```
python3 vol.py -f memdump.raw windows.vadyarascan --yara-rules "5gw4d"
```

6.6 Comparative Study: Volatility 2 vs. Volatility 3

The table below summarises the key differences observed between legacy Volatility 2.x and Volatility 3.0 during this lab exercise.

Feature / Criterion	Volatility 2.x (Legacy)	Volatility 3.0
Python Version	Python 2.7 (EOL). Must use python2 or vol.py wrapper.	Python 3.6+ (actively maintained). Fully modern stack.
Profile Selection	Manual. Analyst must specify --profile= (e.g., Win10x64_19041). Requires imageinfo first.	Automatic. Detects OS version from kernel symbols. No --profile flag needed.
Symbol Files	Uses hard-coded Python profile classes. Limited to profiles shipped with the tool.	Uses JSON Symbol Tables (.json.xz). Can auto-download or be generated with dwarf2json.
Plugin Naming	Short names: pslist, netscan, malfind, cmdscan	OS-prefixed: windows.pslist, windows.netstat, windows.malfind
YARA Integration	Optional; requires separate yara-python dependency and explicit --yara-file flag.	Built-in vadyarascan plugin. Supports inline rules via --yara-rules string argument.
Output Format	Plain text; limited to --output=text or --output=dot. Inconsistent across plugins.	Supports --output=text, --output=csv, --output=json for easy downstream processing.
Performance	Generally slower; single-threaded Python 2 execution.	Faster; Python 3 optimizations and improved layering framework enable better speed.
Plugin Ecosystem	Mature; large library of community plugins (200+). Well-documented online.	Growing; core plugins well-maintained. Community ecosystem still catching up to Volatility 2.
Windows 11 / Latest OS Support	Limited; profiles must be manually created for very recent OS builds. Not officially supported.	Excellent; symbol table generation is automated via Microsoft's PDB server. Supports latest builds.
Recommended Use Case	Legacy investigations; CTFs with older OS dumps; when a specific community plugin only exists in V2.	All modern investigations; preferred for professional forensic workflows; production environments.